

u1-quantum_mechanics

n01-foundational_concepts

These notes were made with Gemini after scanning my handwritten notes.

The Failure of Classical Physics

Classical physics, based on **Newtonian mechanics** and **Maxwell's electromagnetism**, was successful at explaining phenomena for macroscopic objects and at slow speeds. However, it failed to explain several key phenomena:

- **Blackbody Radiation:** The predicted energy distribution from a blackbody did not match experimental results, leading to the **ultraviolet catastrophe**.
- **Photoelectric Effect:** The classical prediction that electrons would be emitted from a metal surface after a delay, and with a kinetic energy dependent on light intensity, was incorrect.
- **Compton Effect:** The scattering of photons by charged particles could not be explained by classical wave theory.
- **Atomic Stability:** Classical physics predicted that orbiting electrons would continuously radiate energy and spiral into the nucleus, causing atoms to collapse.
- **Specific Heat of Solids:** Classical theories couldn't accurately predict how the specific heat of solids changes with temperature.

Blackbody Radiation

A blackbody is an idealized object that absorbs all incident radiation and acts as a perfect emitter.

To analyze the radiation from a blackbody, a hollow cavity with a small hole is used. The radiation inside the cavity is in thermal equilibrium.

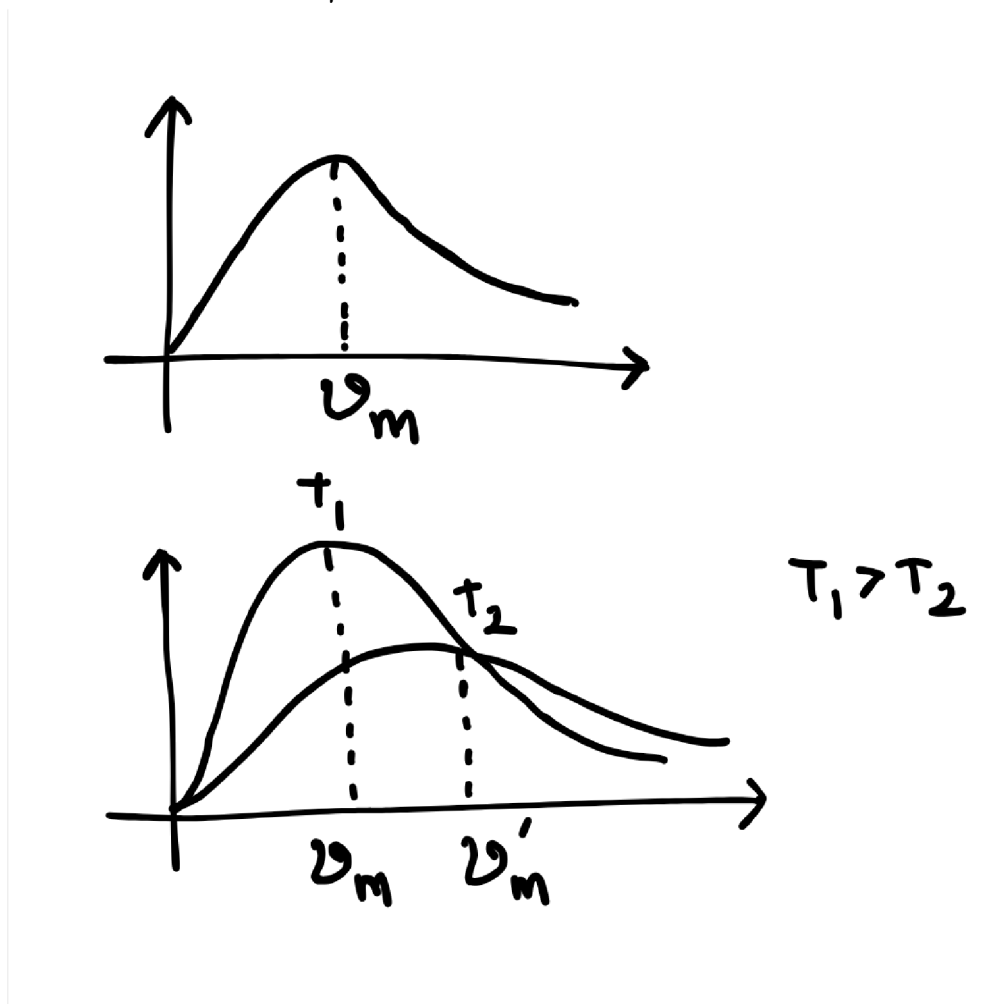
- **Rayleigh-Jeans Law:** This classical law attempted to describe the energy density of blackbody radiation at a given frequency and temperature.
 - It was based on the assumption that standing waves within the cavity are equivalent to harmonic oscillators.
 - The formula derived was

$$u(\nu, T) = \frac{8\pi\nu^2 kT}{c^3}$$

- **Problem:** This law worked only for **low frequencies**. At high frequencies, the formula predicted an infinite energy density, which contradicted experimental observations. This failure is known as the **ultraviolet catastrophe**.

To explain the spectral variation of blackbody radiation, many experiments have been conducted and concluded that at thermal equilibrium:

- energy distribution is uniform and continuous
- increases with frequency and reaches a maximum value at a particular frequency (ν_m) and then decreases
- for a particular frequency, energy density doesn't depend on the chemical composition of the object or shape but depends on the temperature of the inner walls
- ν_m increases as temperature increases



- **Planck's Hypothesis:** In 1900, Max Planck proposed a revolutionary idea to solve the ultraviolet catastrophe.
 - Energy exchange between matter and radiation is **discrete** (quantized).
 - The energy of a radiation oscillator must be an integral multiple of a fundamental unit, or quantum, given by $E = nh\nu$, where $n = 0, 1, 2, \dots$
 - Using this hypothesis, Planck derived a new formula for the energy density of blackbody radiation:

$$u(\nu, T) = \frac{8\pi h\nu^3}{c^3} \cdot \frac{1}{e^{h\nu/kT} - 1}$$

- This formula accurately matched the experimental data across all frequencies.
- **At low frequencies** ($h\nu \ll kT$), the formula reduces to the Rayleigh-Jeans law.
- **At high frequencies** ($h\nu \gg kT$), the denominator becomes very large, causing the energy density to approach zero, solving the ultraviolet catastrophe.
- **Wien's Law:** An earlier empirical law that worked well for high frequencies.

$$\lambda_m T = \text{constant}$$

- **Stefan-Boltzmann Law:** States that the total energy radiated per unit surface area of a blackbody is directly proportional to the fourth power of its absolute temperature:

$$E = T^4$$

Photoelectric Effect

This phenomenon provided strong experimental evidence for the **particle nature** of light.

- **The Phenomenon:** When light shines on a metal surface, electrons are emitted. This is the **photoelectric effect**.
- **Classical Predictions vs. Experimental Results:**
 - **Classical:** Predicted that electron emission would depend on the intensity of light and have a time delay, as energy from the wave would build up.
 - **Experimental:** Showed that electrons are emitted **instantly** if the light's frequency is above a certain **threshold frequency** regardless of intensity.
 - **Classical:** Predicted that the kinetic energy of the emitted electrons would increase with light intensity.
 - **Experimental:** Showed that the **kinetic energy** of the emitted electrons depends on the **frequency** of the light, not the intensity. The intensity of the light only affects the number of electrons emitted (the photoelectric current).
- **Einstein's Explanation (1905):**
 - Building on Planck's work, Einstein proposed that light itself is composed of discrete energy packets called **photons**.

- The energy of a single photon is given by $E = h\nu$.
- One photon interacts with one electron.
- The energy of the photon ($h\nu$) is used to overcome the binding energy of the electron (the **work function**, W) and the remaining energy becomes the electron's kinetic energy (K).
- The **photoelectric equation** is:

$$h\nu = W + K$$

- The photoelectric effect is definitive proof that light behaves as a particle.

$$K_{max} = eV_0$$

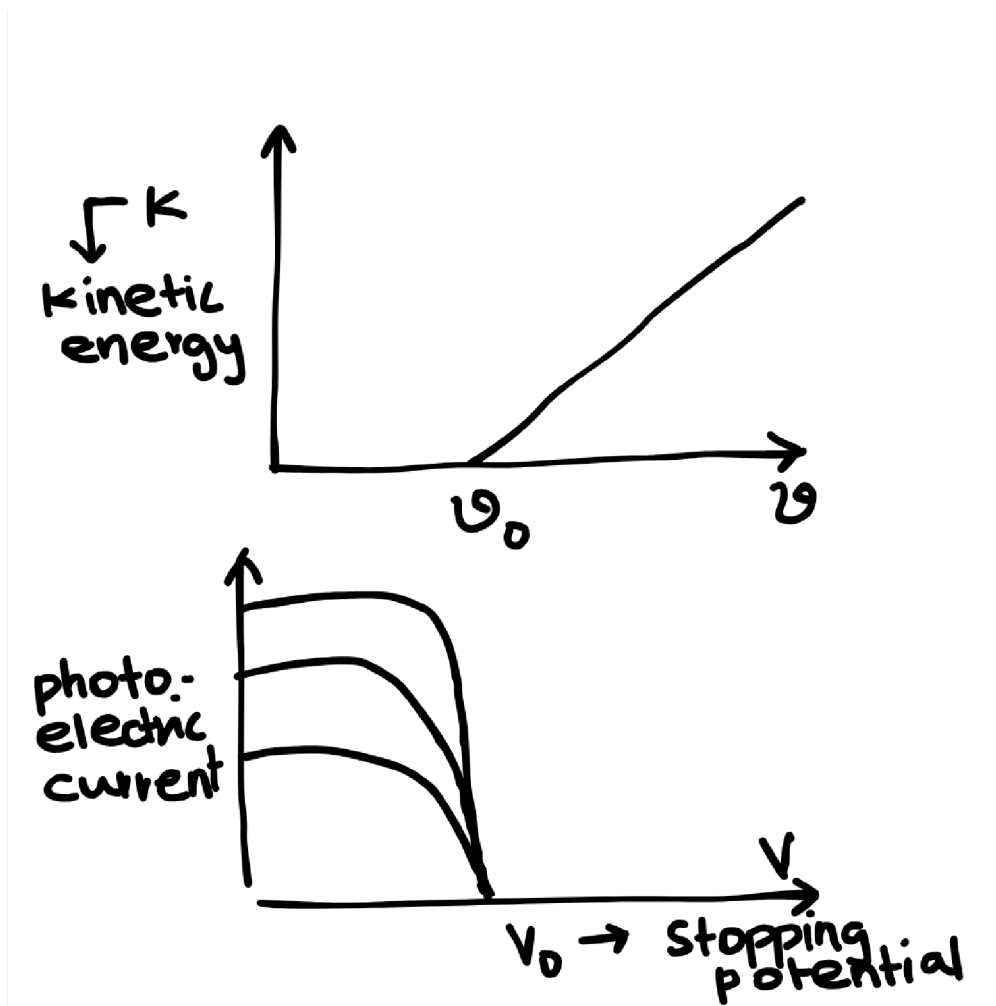
Key Figures

- **Niels Bohr**: Proposed that electrons reside in specific, quantized orbits around the nucleus, explaining atomic stability.
- **De Broglie**: Hypothesized that every particle has a wave nature, leading to the concept of wave-particle duality
- **Schrödinger**: Developed the wave formalism of quantum mechanics.
- **Dirac**: Developed a relativistic version of quantum mechanics.

nO2-photoelectric_effect

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$$h\nu = \phi + K$$

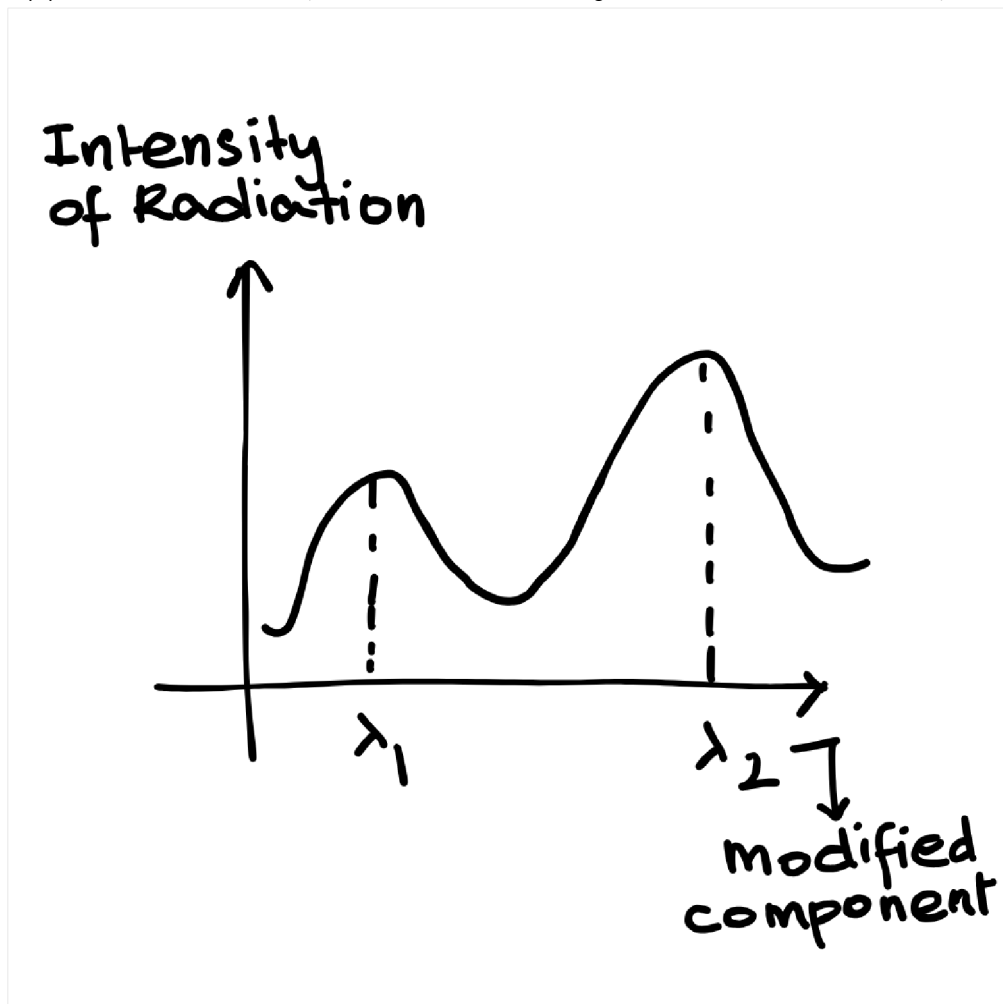
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$$K_{max} = eV_0$$

n03-compton's_shift

Importance

- It provided the experimental evidence for the particle nature of radiation.
- It conclusively demonstrated that the description of a **photon** can be applicable not only to the visible region but also to x-rays.



Definition:

When x-rays (consisting of photons) are incident on a target material, the photon will be scattered in two ways:

- **Scattering wavelength** is the same as the incident wavelength. This is known as the **unmodified component**.

- **Scattering wavelength** is different (longer side) from the incident wavelength. This is the **modified component** of the radiation.
- The difference between these two wavelengths is called the **Compton Shift** or **Compton Scattering**. It's represented by the formula:

$$|\lambda_2 - \lambda_1|$$

where λ_1 is the incident wavelength and λ_2 is the scattered wavelength.

Therefore the phenomenon of change in wavelength upon the incidence of x-rays is called **Compton scattering**.

Note:

- **Bounded electrons** are responsible for **Rayleigh scattering**.
- **Free electrons** are responsible for **Compton shift**.

nO4-wave-particle__duality

The experiments that give direct evidence for the **wave nature of light** are:

- Interference
- Diffraction
- Polarisation

The experiments that give direct evidence for the **particle nature of light** are:

- Black body radiation
- Photoelectric effect
- Compton scattering

Light should act like a particle at one time and like a wave at another, depending on the situation. It never acts as both at the same time.

De-Broglie Hypothesis

In 1924, **De-Broglie** proposed the wave-nature of matter based on the following observations:

- Nature is composed of matter and radiation, and since they are forms of energy, one can be transferred to the other.
- Since nature always likes symmetry, matter also exhibits wave nature like radiation is having **wave-matter particle duality**.
- He proposed that every moving particle is associated with a wave having a wavelength λ :

$$\lambda = \frac{h}{mv}$$

De-Broglie Wavelength (λ)

- If a particle is having kinetic energy E ($E = \frac{p^2}{2m}$), then the wavelength associated with it is:

$$\lambda = \frac{h}{\sqrt{2mE}}$$

- If an electron is accelerated to a potential of ' V ', then:

$$\lambda = \frac{h}{\sqrt{2meV}} = \frac{12.28}{\sqrt{V}} \text{ (volts)}$$

- If the particle is in thermal equilibrium, then:

$$\lambda = \frac{h}{\sqrt{3mkT}}$$

Properties of Matter Waves

- The wave associated with a moving particle is called a **matter wave**.
- Matter wave is **independent of charge** and is **different from the electromagnetic wave**.
- Light rays or particles possess longer wavelength.
- They exhibit a **diffraction pattern** like other waves.
- Slow moving particles have **longer wavelength**.
- Matter waves can **propagate through vacuum** and they don't need any medium to propagate.
- Matter waves can have the velocity **greater than the velocity of light**.

Phase Velocity of de-Broglie Wave (V_p)

Phase velocity, V_p , is the velocity of the planes having constant phase.

$$V_p = \nu\lambda \rightarrow (1)$$

- Where $\nu \rightarrow$ frequency and $\lambda \rightarrow$ wavelength of a de-Broglie wave.

We know that:

$$E = h\nu \rightarrow \nu = \frac{E}{h}$$

$$\nu = \frac{E}{h} \quad \text{and} \quad \lambda = \frac{h}{p} \rightarrow (2)$$

Substituting ν and λ in (1),

$$V_p = \frac{E}{h} \times \frac{h}{p}$$

$$V_p = \frac{E}{p} \rightarrow (3)$$

Case-1: Non-Relativistic Particle

If the particle is **non-relativistic**, then

$$E = mc^2$$

$$P = mv$$

Where m and v are **mass and velocity** of the moving particle.

$$V_p = \frac{mc^2}{mv} = \frac{c^2}{v} \rightarrow (4)$$

From equation (4), it is evident that V_p is **always greater than the velocity of light** since $v \ll c$.

Case-2: Relativistic Particle

If the particle is **relativistic**, then

$$E = \sqrt{m_0^2 c^4 + p^2 c^2}$$

Substituting E in equation (3):

$$V_p = \frac{\sqrt{m_0^2 c^4 + p^2 c^2}}{p}$$

$$V_p = \left(\frac{p^2 c^2 + m_0^2 c^4}{p^2} \right)^{1/2}$$

$$V_p = c \left(1 + \frac{m_0^2 c^2}{p^2} \right)^{1/2}$$

We know that $p = h\lambda$. Substituting this:

$$V_p = c \left(1 + \frac{m_0^2 c^2 \lambda^2}{h^2} \right)^{1/2} \rightarrow (5)$$

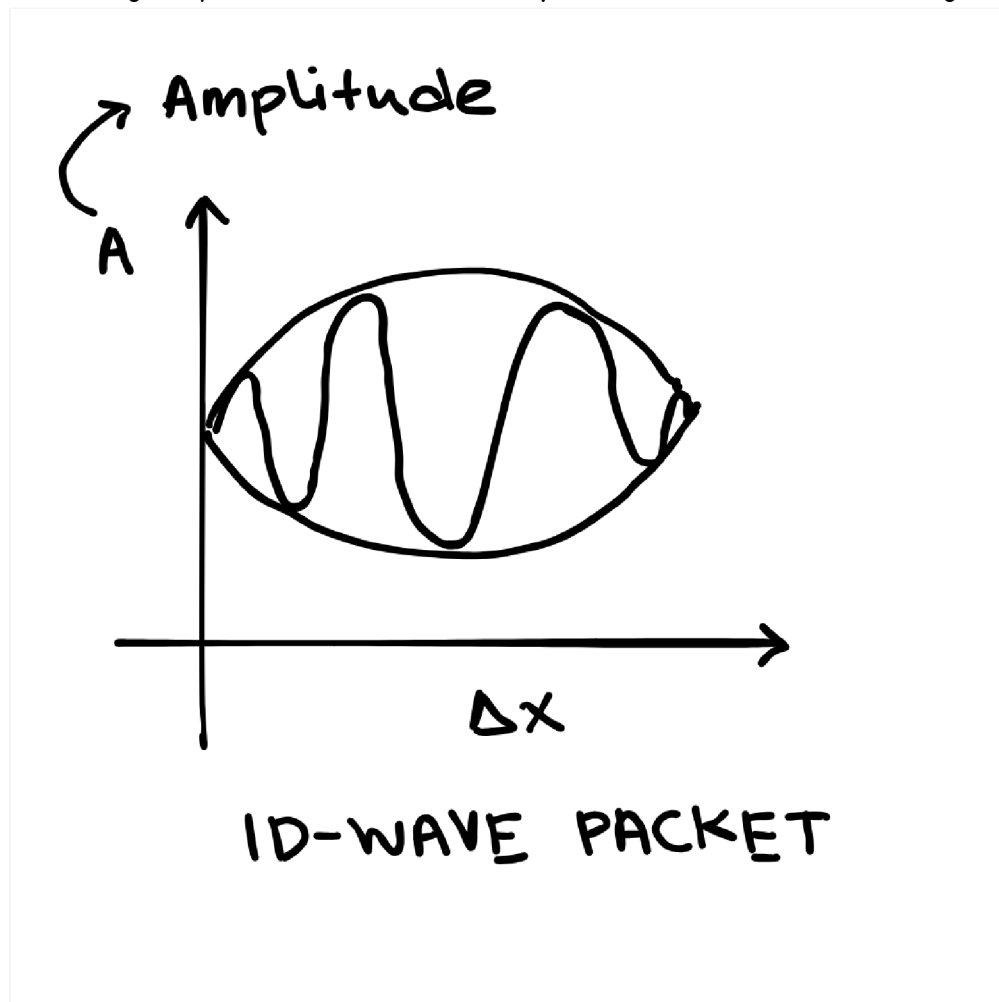
In the above equation (5), $\frac{m_0^2 c^2 \lambda^2}{h^2}$ is **always positive**.

Thus, $V_p > c$.

Therefore, the velocity of the de-Broglie wave is **always greater than the velocity of light**.

Wave Packet and Group Velocity

Note: Since V_p is always greater than (refer equation 4), the wave propagates faster than the particle, leaving the particle far behind. This problem was resolved by **Schrödinger** by postulating that every moving particle is associated with a group of waves i.e. **wave packet** rather than a single wave.



- A **wave packet** is comprised of a number of waves in different wavelengths and amplitudes with phases and different wave numbers chosen in such a way that they **interfere constructively** in a particular region where the particle can exist and at other regions **destructive interference** takes place so that **amplitude is zero**.
- The velocity of the wave packet is called the **group velocity** (V_g) and the velocity of individual waves that comprise the wave packet is called the **phase velocity** (V_p).

Expression for Group Velocity

Let us consider a wave packet comprising of two waves:

$$y_1 = A \cos(t - kx)$$

$$y_2 = A \cos((+ \Delta)t - (k + \Delta k)x)$$

having same amplitude but different angular frequency (Δ) and wave propagation constant (Δk) by Δ and Δk .

$$y = y_1 + y_2$$

$$y = A \cos(t - kx) + \cos((+ \Delta)t - (k + \Delta k)x)$$

$$y = 2A \cos\left(\frac{(t - kx) + ((+ \Delta)t - (k + \Delta k)x)}{2}\right) \cos\left(\frac{(t - kx) - ((+ \Delta)t - (k + \Delta k)x)}{2}\right)$$

$$y = 2A \cos\left(\left(+ \frac{\Delta}{2}\right)t - \left(k + \frac{\Delta k}{2}\right)x\right) \cos\left(\frac{\Delta}{2}t - \frac{\Delta k}{2}x\right)$$

Since (Δ) and (Δk) are very small compared to t and kx

$$y \approx 2A \cos(t - kx) \cos\left(\frac{\Delta}{2}t - \frac{\Delta k}{2}x\right)$$

Relation between Group velocity and Phase velocity

i) The wave traveling with angular frequency and wave propagation constant k with velocity:

- **Phase Velocity:**

$$V_p = \frac{\omega}{k}$$

When the Δ and Δk are very small then V_g is determined by

- **Group Velocity:**

$$V_g = \frac{d\omega}{dk} = \frac{\Delta\omega}{\Delta k}$$

ii) We know that

$$\omega = V_p k$$

Differentiating with respect to k :

$$\frac{d\omega}{dk} = V_p \frac{dk}{dk} + k \frac{dV_p}{dk}$$

$$V_g = V_p + k \frac{dV_p}{dk}$$

We know that $k = \frac{2\pi}{\lambda}$, then

$$\frac{dk}{d\lambda} = \frac{-2\pi}{\lambda^2}$$

Now,

$$\frac{dV_p}{dk} = \frac{dV_p}{d\lambda} \frac{d\lambda}{dk}$$

$$\frac{dV_p}{dk} = \frac{dV_p}{d\lambda} \left(\frac{-\lambda^2}{2\pi} \right)$$

Substituting the relationship between dk and $d\lambda$:

$$V_g = V_p + k \frac{dV_p}{d\lambda} \left(\frac{-\lambda^2}{2\pi} \right)$$

$$V_g = V_p + \left(\frac{2\pi}{\lambda} \right) \frac{dV_p}{d\lambda} \left(\frac{-\lambda^2}{2\pi} \right)$$

$$V_g = V_p - \lambda \frac{dV_p}{d\lambda}$$

Conditions for non-dispersive and dispersive medium

- **Non-dispersive medium** ($V_p = \text{constant}$): If all the waves are travelling with same velocity for all wavelengths.

$$\frac{dV_p}{d\lambda} = 0$$

$$V_g = V_p$$

- **Dispersive medium**: The different wavelengths travel with different speeds, V_g is always less than V_p .

Derivation of Group Velocity in terms of Momentum

Let us consider a particle of mass 'm' moving with velocity V is associated with a de Broglie wave of velocity V_p .

- **Phase velocity**:

$$V_p = \nu \lambda$$

- **De Broglie Wavelength**:

$$\lambda = \frac{h}{p}$$

- **Energy**:

$$E = h\nu =$$

- **Wave Number**:

$$k = \frac{2\pi}{\lambda} = \frac{p}{h}$$

From $E = h\nu$ $dE = h d\nu$.

From $k = \frac{2\pi}{\lambda}$ $dk = -\frac{2\pi}{\lambda^2} d\lambda$.

Dividing the expressions for dE and dk :

$$\frac{dE}{dp} = \frac{d}{dk} = \frac{d}{dk}$$

- **Group Velocity:**

$$V_g = \frac{d}{dk} = \frac{dE}{dp}$$

For a non-relativistic particle:

- **Energy:**

$$E = \frac{p^2}{2m}$$

$$V_g = \frac{dE}{dp} = \frac{d}{dp} \left(\frac{p^2}{2m} \right) = \frac{2p}{2m} = \frac{p}{m}$$

Since **momentum** $p = mV$:

$$V_g = V$$

Therefore, **Group velocity is always equal to the velocity the particle associated with a de Broglie wave.**

Relativistic Correction for Group Velocity

Considering the relativistic correction for the energy and momentum:

- **Relativistic Energy:**

$$E = \frac{m_0 c^2}{\sqrt{1 - \frac{V^2}{c^2}}}$$

- **Relativistic Momentum:**

$$p = \frac{m_0 V}{\sqrt{1 - \frac{V^2}{c^2}}}$$

We use $V_g = \frac{d}{dk}$.

$$= \frac{2\pi}{h} E = \frac{2\pi}{h} \frac{m_0 c^2}{\sqrt{1 - \frac{V^2}{c^2}}}$$

$$d = \frac{2\pi m_0 V}{h(1 - \frac{V^2}{c^2})^{3/2}} dV \quad \dots (5)$$

$$k = \frac{2\pi}{h} p = \frac{2\pi}{h} \frac{m_0 V}{\sqrt{1 - \frac{V^2}{c^2}}}$$

$$dk = \frac{2\pi m_0}{h(1 - \frac{V^2}{c^2})^{3/2}} dV \quad \dots (6)$$

The group velocity is $V_g = \frac{d}{dk}$

Substituting equations (5) and (6) into the expression for V_g :

$$V_g = \frac{\frac{2\pi m_0 V}{h(1-V^2/c^2)^{3/2}} dV}{\frac{2\pi m_0}{h(1-V^2/c^2)^{3/2}} dV}$$

$$V_g = V$$

Therefore, Group velocity is always equal to the velocity of the particle associated with a de Broglie wave.

n05-schrödinger_wave_equation

Wave Function

The quantity whose variations makes the Matter waves is called wave function and is denoted by ψ .

ψ is always a function of $\vec{r}$ and t and exists at all points in space.

Schrödinger Wave Equation

The wave equation which describes the particle's nature in terms of periodic displacement in space ($\vec{r}$) and time (t) is called the **Schrödinger wave equation**.

Time-Independent Schrödinger Wave Equation

Let's consider a group of **stationary waves** associated with a moving particle represented by $(\vec{r}, t)$.

The **3D wave equation** which represents the behavior of a particle classically is given by:

$$\nabla^2 = \frac{1}{V^2} \frac{\partial^2}{\partial t^2} \quad \dots (1)$$

where

$$\nabla^2 = (\nabla \cdot \nabla) = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right)$$

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

and V = velocity of the particle.

The solution of equation (1) is given by considering $(\vec{r}, t)$ at $t = 0$ as:

$$(\vec{r}, t) = \psi_0(\vec{r}) e^{-it} \quad \dots (2)$$

where $\psi_0(\vec{r})$ is the **amplitude** of the wave at $t = 0$.

Differentiating the above equation with respect to t twice we get:

$$\frac{\partial}{\partial t} = {}_0(\vec{r})(-i)e^{-it}$$

$$\frac{\partial^2}{\partial t^2} = {}_0(\vec{r})(-i)^2 e^{-it}$$

$$\frac{\partial^2}{\partial t^2} = -^2{}_0(\vec{r})e^{-it}$$

$$\frac{\partial^2}{\partial t^2} = -^2(\vec{r}, t) \quad (\text{fro 2})$$

Substituting the value of $\frac{\partial^2}{\partial t^2}$ in equation (1):

$$\nabla^2(\vec{r}, t) = \frac{-^2}{V^2}(\vec{r}, t)$$

$$\nabla^2(\vec{r}, t) = -\frac{4\pi^2\nu^2}{V^2}(\vec{r}, t)$$

using the wave speed relationship $V = \nu\lambda$, or $\frac{\nu}{V} = \frac{1}{\lambda}$ (where $= 2\pi\nu$):

$$\nabla^2(\vec{r}, t) = -\frac{4\pi^2}{\lambda^2}(\vec{r}, t)$$

$$\nabla^2(\vec{r}, t) + \frac{4\pi^2}{\lambda^2}(\vec{r}, t) = 0$$

Incorporating De Broglie and Energy Concepts

Introducing the **de Broglie concept** for the waves we have $\lambda = \frac{h}{p}$.

Substituting this into the equation:

$$\nabla^2(\vec{r}, t) + \frac{4\pi^2 p^2}{h^2}(\vec{r}, t) = 0 \quad \dots (3)$$

If E and V represent **total** and **potential energy**:

Kinetic Energy (K.E.) $= E - V$

We also know that K.E. $= \frac{p^2}{2m}$

$$\frac{p^2}{2m} = E - V \quad \left(\dots = \frac{p^2}{2m} \right)$$

$$p^2 = 2m(E - V)$$

Using the **reduced Planck constant** $= \frac{h}{2\pi}$ or $h = 2\pi$:

Substituting the values of p^2 and h in equation (3):

$$\nabla^2(\vec{r}, t) + \frac{4\pi^2 \cdot 2m(E - V)}{(2\pi)^2}(\vec{r}, t) = 0$$

$$\nabla^2(\vec{r}, t) + \frac{4\pi^2 \cdot 2m(E - V)}{4\pi^2}(\vec{r}, t) = 0$$

$$\nabla^2(\vec{r}, t) + \frac{2m(E - V)}{2}(\vec{r}, t) = 0$$

This is the Time-Independent Schrödinger Wave Equation.

Note: For a free particle ($V = 0$) Schrödinger's time independent equation is given by:

$$\nabla^2 + \frac{2mE}{2} = 0$$

Time Dependent Wave Equation

Schrödinger time dependent wave equation can be obtained by eliminating E from equation (4).

$$\nabla^2 = \frac{-2m}{2}(E - V)$$

$$-\frac{2}{2m}\nabla^2 = E - V$$

$$E = V - \frac{2}{2m}\nabla^2 \quad \text{--- (5)}$$

Differentiating eq: (2) w.r.t t , we get,

$$\frac{\partial}{\partial t} = (\vec{r})(-i)e^{-it}$$

$$\frac{\partial}{\partial t} = -i(\vec{r})e^{-it}$$

$$\frac{\partial}{\partial t} = -i\left(\frac{E}{2}\right) \quad (E =)$$

$$\frac{\partial}{\partial t} = \frac{-iE}{2}$$

$$E = i\frac{\partial}{\partial t} \quad \text{--- (6)}$$

Substituting value of E from eq (6) to eq (5)

$$i\frac{\partial}{\partial t} = V - \frac{2}{2m}\nabla^2$$

$$\left(V - \frac{2}{2m}\nabla^2\right) = \left(i\frac{\partial}{\partial t}\right) \quad \text{--- (7)}$$

Since the above equation involves variation w.r.t time, this is called **time dependent Schrödinger wave equation**.

Eq (7) can also be written as

$$\hat{H} = E \quad \text{--- (8)}$$

where $\hat{H}$ is **Hamiltonian Operator**, where $\hat{H} = V - \frac{\hbar^2}{2m} \nabla^2$

$E = i \hbar \frac{\partial}{\partial t}$ (**Energy Operator**).

It gives probability of particle.

n06-max_born_approximations

- **Max Born** postulated that since $|\psi(x)|^2$ is the probability of finding the particle at x and t .
- Suppose a particle is moving along x -axis with a matter wave having wave function as shown in the figure, then $|\psi(x)|^2 dx$ is the probability between x and dx . $\int_{-\infty}^{\infty} |\psi(x)|^2 dx$ is probability of entire region.

Note: $P = \int_{-\infty}^{\infty} |\psi(x)|^2 dx$ is always between 0 and 1.

Admissible Wave Functions

Admissible wave functions a.k.a. **well-behaved wave functions** are the wave functions which satisfy the following conditions:

- It must be a solution of the **Schrödinger wave equation**.
- It must be **continuous**.
- It must be **finite** for all values for all values of x, y, z at an instant of time t .
- It must be **single-valued**.
- It must **vanish at boundaries**.
- It must be **normalized**, i.e. $\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1$.
- It must be **analytical**, i.e. **first-order derivative must be continuous**

n07-applications_of_schrödinger_wave_equation

Application of Time-Independent S.E.: Infinite Square Well (1D Box)

Consider a particle of mass m moving back and forth in a rigid box (fig 1). Let the particle be moving along the x -axis having boundaries $x = 0$ and $x = a$. When the particle collides with the walls of the box, no energy will be lost and inside the box, V will be uniform.

This can be treated as a particle moving in infinite square well potential whose potential energy V is defined as:

$$V(x) = \begin{cases} 0 & 0 < x < a \\ \infty & x = 0 \text{ and } x = a \end{cases} \quad (1)$$

Since $V(x) = \infty$ outside the boundaries, the wave function of the particle is zero when $x = 0$ or $x = a$.

For Region II ($0 < x < \infty$):

Substituting $V = 0$ in the T.I.S.E.:

$$\frac{d^2}{dx^2} + \frac{2m}{2}(E - 0) = 0$$

$$\frac{d^2}{dx^2} + \frac{2mE}{2} = 0$$

Let

$$k^2 = \frac{2mE}{2} \quad \dots (2)$$

. The equation is:

$$- + = \quad (3)$$

The solution of (3) is given by:

$$\psi(x) = A \sin(kx) + B \cos(kx)$$

where A and B are constants to be determined.

Applying the boundary conditions, the wave function must vanish at $x = 0$ and $x = \infty$:

I. At $x = 0$, $\psi(0) = 0$:

$$\psi(0) = A \sin(0) + B \cos(0) = 0$$

$$0 + B \cdot 1 = 0 \implies B = 0 \quad (4)$$

Thus, the wave function is:

$$\psi(x) = A \sin(kx)$$

II. At $x = \infty$, $\psi(\infty) = 0$:

$$A \sin(k\infty) + B \cos(k\infty) = 0 \quad (4)$$

Using $B = 0$ (from (4)):

$$A \sin(k\infty) = 0$$

A should not be zero (because ψ becomes 0 everywhere).

Therefore, $\sin(k\infty) = 0$:

$$k = n\pi$$

where $n = 1, 2, 3, \dots$

The wave number is quantized:

$$k = \frac{n\pi}{a} \quad (5)$$

$$\psi(x) = \sin\left(\frac{n\pi x}{a}\right) \quad (8)$$

NOTE: $n = 0$ makes the wave function $\psi(x)$ to be zero everywhere.

Energy Eigenvalues

From eq (7) ($k^2 = \frac{2mE}{\hbar^2}$) and eq (8) ($k = \frac{n\pi}{a}$):

$$\left(\frac{n\pi}{a}\right)^2 = \frac{2mE}{\hbar^2}$$

$$\frac{n^2\pi^2}{a^2} = \frac{2mE}{\hbar^2(4\pi^2)}$$

$$\frac{n^2\pi^2}{a^2} = \frac{8m\pi^2 E}{\hbar^2}$$

Solving for E :

$$E = \frac{\hbar^2 n^2 \pi^2}{8ma^2}$$

From the above equation

It is clear that

- the minimum energy of the system is given by putting $n = 1$ i.e.

$$E_1 = \frac{\hbar^2 \pi^2}{8ma^2}$$

this is called **zero point energy**

- The energies of higher levels of the system are given by

$$E_2 = (2)^2 E_1 = 4E_1$$

$$E_{n+1} - E_n = (2n + 1)E_1$$

- the spacing between the n^{th} level and $(n + 1)^{th}$ level is given by

$$E_{n+1} - E_n = (2n + 1)E_1$$

The above energy values are called **eigenvalues** and the corresponding wave function is known as **eigenfunction**

$$\int_0^1 P(x) dx = 1$$

$$\int_0^1 A^2 \sin^2(x) dx = 1$$

$$\int_0^1 A^2 \sin^2\left(\frac{n\pi x}{2}\right) dx = 1$$

$$\frac{A^2}{2} \int_0^1 (1 - \cos(2n\pi x)) dx = 1$$

$$\frac{A^2}{2} \left\{ (-0) - \left(\frac{-\sin(2n\pi)}{2n\pi} + 0 \right) \right\} = 1$$

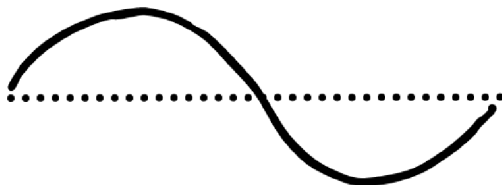
$$\frac{A^2}{2} = 1$$

$$A = \sqrt{2}$$

$n=1$



$n=2$



$n=3$



$\psi_n(x)$

From eqn 10, it is clear that ψ_1 has

- 2 nodes when $x = 0$ and $x = 1$
 - ψ_2 has 3 nodes when $x = 0, 0.5$ and $x = 1$
 - ψ_3 has 4 nodes when $x = 0, 0.33, 0.66, 1$
- Similarly ψ_n has $(n + 1)$ nodes

- probability of finding a particle over a small distance dx at x is given by

$$P(x)dx = |\psi_n(x)|^2 dx$$

$$P(x) = \frac{2}{L} \sin^2\left(\frac{n\pi x}{L}\right)$$

Thus the probability density for 1D motion is

$$P = \frac{2}{L} \sin^2\left(\frac{n\pi x}{L}\right)$$

the probability is maximum when

$$\frac{n\pi x}{L} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$$

$$x = \frac{L}{2n}, \frac{3L}{2n}, \frac{5L}{2n}, \dots$$

Therefore

$$\psi_n = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \quad \dots (10)$$

$n=1$



$n=2$



$n=3$



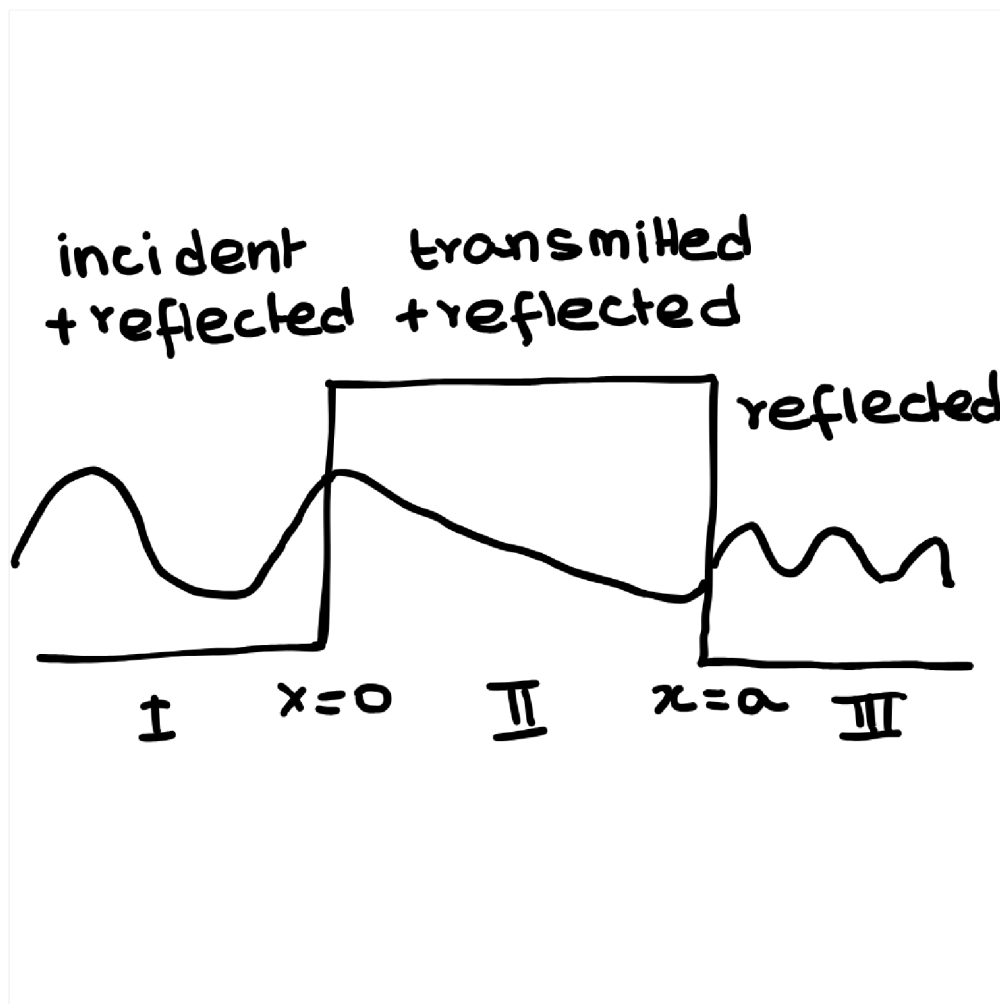
$$P = \psi_n^2(x)$$

- As shown in the figure, at $n = 1$, the most probable position is at $x = \frac{L}{2}$
- for $n = 2$, the most probable position is at $x = \frac{L}{4}, \frac{3L}{4}$

Numerical

1. When electron is confined in a 1D potential well having width $= 8 \times 10^{-10}$, Find the kinetic energy of electron when it is in the **ground state**.
2. Electron is bound in a 1D box having size of 4×10^{-10} .
3. Electron is bound in 1D infinite well having a width of 1×10^{-10} . Find the energy values in the ground state and 1st 2nd excited states.

n08-quantum_tunneling



$$V(x) = \begin{cases} 0 & x < 0 \\ V_0 & 0 < x < a \\ 0 & x > a \end{cases}$$

- The ability of a particle to cross the barrier, even if its energy is lower than that of the height of the barrier is called quantum tunneling.
- the **transmission coefficient** is given by

$$T = e^{-2\kappa a}$$

- where

$$k_1 = \sqrt{\frac{2m(-E)}{\hbar^2}}$$

1. The points $x = 0$ and $x = a$ are called **turning points**.
2. **Quantum tunneling** depends on k_1 and k_2 .
3. In region 2, the wave decays exponentially and reinforces before it comes to zero.
4. So in region 3, even though the amplitude is small, the particle never loses energy.